Muzhe Wu

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Objectives

I study how people, individually and collectively, interact with AI systems and design representations and tools towards more desirable futures. My past work focused on augmenting people's capabilities in the physical world (e.g., task guidance/learning, navigation). My current work seeks to broaden the conception of "intelligence" to embrace human ambiguity and subjectivity (e.g., values, creativity).

Education

University of Michigan (UMich) — Aug 2025 – Present
Ph.D. Computer Science (GPA: 4.00/4.00) — Ann Arbor, MI

Carnegie Mellon University (CMU), HCII — Aug 2023 – Apr 2025
M.S. Educational Technology & Applied Learning Sciences (QPA: 4.16/4.33) — Pittsburgh, PA

Shanghai Jiao Tong University (SJTU) — Sep 2019 – Aug 2021, May – Aug 2023
B.S. Electrical and Computer Engineering (GPA: 3.70/4.00) — Shanghai, China

University of Michigan — Aug 2021 – Apr 2023
B.S. Computer Science (GPA: 3.95/4.00) — Ann Arbor, MI

Full-Paper Publications

*: equal contributions.

[C.4] **Muzhe Wu***, Yanzhi Zhao*, Shuyi Han, Michael Xieyang Liu, Hong Shen. 2024. AI LEGO: Scaffolding Cross-Functional Collaboration in Industrial Responsible AI Practices during Early Design Stages. *ACM Conference on Computer-Supported Cooperative Work and Social Computing (CSCW 2026)*. [DOI]

[C.3] **Muzhe Wu***, Haocheng Ren*, Gregory Croisdale, Anhong Guo, Xu Wang. 2023. Rubikon: Intelligent Tutoring for Rubik's Cube Learning Through AR-enabled Physical Task Reconfiguration. *ACM Conference on Designing Interactive Systems (DIS 2025)*. [DOI] [Video]

[C.2] **Muzhe Wu***, Yi Fei Cheng*, David Lindlbauer. 2024. New Ears: An Exploratory Study of Audio Interaction Techniques for Performing Search in a Virtual Reality Environment. *IEEE International Symposium on Mixed and Augmented Reality (ISMAR 2024)*. [DOI] [Video]

[C.1] Ying-Jui Tseng, Gautam Yadav, Xinying Hou, **Muzhe Wu**, Yun-Shuo Chou, Claire Che Chen, Chia-Chia Wu, Shi-Gang Chen, Yi-Jo Lin, Guanze Liao, Kenneth R. Koedinger. 2024. ActiveAI: The Effectiveness of an Interactive Tutoring System in Developing K-12 AI Literacy. *European Conference on Technology Enhanced Learning (EC-TEL 2024)*. [DOI]

Paper Under Review

[U.1] **Muzhe Wu***, Zuchen Li*, Xu Wang, Anhong Guo. 2026. Generative Tutorial: Towards Live Contextualized Visual Instructions for Physical Tasks. [arXiv]

Selected Experience

EECS 281 Data Structures and Algorithms, UMich — Sep 2026 – Present
Graduate Student Instructor (GSI) — Ann Arbor, MI
Support the 629-student course through lab instruction, office hours, material development, discussion-forum monitoring, and exam grading; coordinate with course staff to deliver effective guidance.

Human-AI Lab & Lifelong Learning Lab, UMich — Aug 2025 – Sep 2026
Graduate Student Research Assistant (Advisor: Anhong Guo & Xu Wang) — Ann Arbor, MI

Developed an intelligent tutoring system for Rubik's Cube learning; developed an AR-based live contextualized visual instruction system with visual generative models for everyday physical tasks.

Augmented Perception Lab, CMU Jan 2024 – Apr 2025
Research Assistant (Advisor: David Lindlbauer) Pittsburgh, PA
Developed novel audio interaction techniques for VR search and navigation tasks; developed a computational UI adaptation method to monitor and balance muscle strain while sitting; measured the trade-off between user performance and agency in target selection tasks under preemptive assistance.

Collective AI Research and Evaluation Lab, CMU Oct 2023 – Sep 2024
Research Assistant (Advisor: Hong Shen) Pittsburgh, PA
Led co-design studies with industry AI practitioners on cross-functional collaboration for early-stage AI risk identification; developed a web-based collaboration tool to assist industry AI practitioners in planning AI system development and identifying unethical design choices.

Jim Team, NVIDIA Jul – Oct 2022
Developer (Advisor: Jim Fan) Remote
Built an interactive retro game simulation environment for RL agent training; enabled GPU acceleration for MineDojo simulation on headless machines.

Honors, Awards and Grants

Weinberg Institute Cognitive Science Graduate Fellowship, UMich ($5,000) Mar 2025
Scholarly Project (formerly GuSH) Funding, CMU Nov 2023
James B. Angell Scholar, UMich Mar 2023
Merit Scholarship, CMU ($7,000) Feb 2023
Dean's Honor List, UMich Dec 2021, Apr, Dec 2022
Best Demo Award, Michigan AI Symposium Nov 2022
University Honors, UMich Dec 2021, Apr 2022
Undergraduate Excellent Scholarship, SJTU (top 10%) Nov 2020, Nov 2021
Meritorious Winner, Mathematical Contest in Modeling (MCM) (top 9.5%) Feb 2021
Silver Medal, University Physics Competition (top 3%) Nov 2020

Services

Reviewer for CHI (4), CHI LBW & Poster (9), ISMAR (1) 2023 – Present
Student Advisor at Wenzhou No. 2 Foreign Language School ($N = 700$) May 2020
Volunteer at Jiangchuan Sunshine Nursing Home, Shanghai, China Oct 2019 – Aug 2020

Relevant Coursework

HCI: Interaction Design, Prototyping Algorithmic Experiences, Interactive Extended Reality, Engineering Interactive Systems, Human-AI Interaction & Systems, Educational Design, HCI Systems
AI/ML: Machine Learning, Natural Language Processing, Deep Learning for Computer Vision, Science for Deep Learning, Machine Learning in Production, AI for Education, Advanced Scalable Systems
Software: Web Systems, Operating Systems, UI Development, Data Structures & Algorithms
Hardware: Circuits & Signals, Logic Design, Semiconductor Devices, Computer Architecture

Skills

Programming Skills: Python, JavaScript, C#, R, C/C++, Java, SQL, Vibe Coding
Frameworks/Libraries: Meta XR SDK, ARKit, React.js, PyTorch, SwiftUI, AWS, Firebase
Tools/Software: Unity, Figma, LaTeX, MTurk, Docker, Adobe Creative Suite, Ableton, Matlab